

Mixed-Precision Arithmetic

Mixed-precision arithmetic means that one algorithm uses more than one floating-point format. The idea is simple. One tries to perform the expensive part of the computation in a cheaper arithmetic and keeps the more accurate format for the numerically sensitive stages. In numerical linear algebra, however, this is not only a matter of implementation. A change of precision changes the perturbation model of the algorithm itself, and therefore changes the map that is actually computed [1] [2].

This chapter introduces the basic concepts needed later for randomized least-squares solvers. In particular, we discuss floating-point representation, rounding, conditioning, and the way mixed precision interacts with algorithmic structure. The main question for the later chapters is the following: if the algorithm already contains approximation through randomized sketching, how much additional perturbation from lower precision can be tolerated?

1.1 Floating-Point Model

Floating-point arithmetic replaces the continuum of real numbers by a finite set of representable values. A normalized floating-point number has the schematic form

$$x = \pm m\beta^e,$$

where β is the base, m is the significand, and e is the exponent. Since most real numbers are not exactly representable, every arithmetic operation involves rounding. Thus finite precision is not a small correction to an exact algorithm. It is the environment in which the algorithm exists.

For rounding to nearest, the standard model writes one basic operation as

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

where u is the unit roundoff of the working format. Lower precision means a larger u , a smaller dynamic range, and therefore a smaller safety margin against cancellation, overflow, and accumulation of rounding errors. The important point is that changing precision changes the size of the perturbation allowed at every arithmetic step. This is why low precision can be attractive computationally and dangerous numerically at the same time [2] [3] [4].

The standard rounding model is useful because it allows us to interpret the computed algorithm as a nearby problem. Once we pass from one scalar operation to inner products, matrix-vector products, or matrix-matrix products, the perturbations are no longer controlled only by u , but also by the dimensions and the algebraic structure of the computation. For least-squares problems, this is especially important. Residuals, orthogonal factorizations, normal equations, and sketching operations all inherit their own precision-dependent perturbations.

1.2 Why Mixed Precision Matters

Mixed precision has become important because current hardware does not treat all precisions equally. Modern CPUs and especially accelerators often provide much higher throughput for half- or single-precision kernels than for double precision, while also reducing memory traffic and storage costs. Therefore, what used to be an implementation detail has become a design question for algorithms themselves [1] [2].

The real appeal, however, is algorithmic. Many linear algebra procedures are not equally sensitive to rounding. Bulk matrix products may tolerate lower precision, whereas factorizations, orthogonalization, residual computations, or correction solves may require a more accurate format. Mixed precision is therefore best understood as a way of deciding where accuracy is needed and where it is not. One does not choose only how much arithmetic is performed, but also where the larger perturbations are allowed to enter the computation.

1.3 Error Sources and Conditioning

In linear algebra computations, several error sources act at the same time. The data may already be perturbed when stored in low precision. Each arithmetic operation contributes a local rounding error. These perturbations accumulate through matrix products, factorizations, and iterative processes. Finally, the whole effect is filtered through conditioning. A well-conditioned problem may absorb the perturbation harmlessly, while an ill-conditioned one may amplify it strongly. Therefore, forward error, backward error, and conditioning must be kept distinct. Mixed precision changes the perturbation introduced by the algorithm, while the problem itself determines how much this perturbation is amplified.

This is especially important for least-squares problems. One must distinguish between the conditioning of the approximation problem itself and the conditioning of the algebra actually used to solve it. The latter may involve normal equations, QR factorizations, augmented systems, or preconditioned iterative refinement. Recent work on mixed-precision least squares shows that lower precision can be effective, but only in conditioning regimes that still allow stable refinement or correction [5] [6] [7] [8].

1.4 Mixed Precision in Linear Algebra Algorithms

The classical example is iterative refinement. One performs the expensive solve in lower precision, computes a more accurate residual in higher precision, and uses this residual to correct the solution. This idea is old, but modern hardware has made it newly important. Foundational work showed that one can often recover double-precision accuracy while exploiting cheaper lower-precision factorization

steps, and more recent GPU implementations show substantial speed gains when tensor-core arithmetic is used carefully [9] [10] [2].

For least-squares problems the situation is more subtle than for square linear systems, because orthogonality, residual quality, and conditioning of augmented formulations all matter. This is why mixed-precision least-squares solvers often combine low-precision bulk work with higher-precision residuals, QR-based corrections, or Krylov refinement steps [5] [6]. The main issue is not only whether the problem can be solved cheaply, but whether the approximation to the projection onto the range of the data operator remains sufficiently accurate.

Recent work extends this picture beyond standard least squares. Mixed-precision FGMRES-based iterative refinement has been developed for weighted least squares, and related augmented-system approaches have been proposed for constrained and generalized least-squares problems [11] [7]. The common lesson is that lower precision is most useful in the bulk linear algebra that constructs an approximate solve or preconditioner, while the residual and correction stages must remain sufficiently accurate to control the final error.

1.5 Connection to Randomized Numerical Linear Algebra

Randomized algorithms add a second approximation before floating-point effects are even considered. A sketching matrix S replaces the original least-squares problem by a reduced one involving SA and Sb . Mixed precision can then enter at several places: in storing the data, in forming the compressed operator, in solving the reduced problem, and in any refinement step that follows.

These stages need not have the same numerical requirements. If the dominant cost lies in forming SA , low-precision matrix multiplication may be attractive. If the reduced problem is ill-conditioned, however, that gain may disappear unless the subsequent solve or residual computation is more accurate. Recent work makes this point explicit. Mixed-precision analysis of sketching-based preconditioners shows that one can choose separate precisions for the sketching and factorization stages and still obtain an effective preconditioner for GMRES-based iterative refinement, provided the perturbations remain compatible with the conditioning of the least-squares problem [12]. On the implementation side, hardware-oriented work on random projection exploits the fact that the random matrix itself may tolerate reduced precision better than the data matrix or the final solve [13].

The guiding view of this thesis is therefore the following. Sketching error and floating-point error should be analyzed together as coupled perturbations of the original least-squares problem. In the sketch-and-solve setting, one asks whether the computed reduced problem still represents the original one faithfully enough. In the sketch-and-precondition setting, one asks whether lower precision preserves

enough of the conditioning improvement for the downstream solver to remain effective.

1.6 Balancing Sources of Inexactness

Mixed precision is most natural when rounding is not the only source of error in the algorithm. Randomized approximation, low-rank truncation, regularization, and even finite-sample modelling choices already introduce tolerances below which additional arithmetic accuracy may not be useful. In such settings, the purpose of precision selection is not to minimize rounding error at all costs. It is to choose an arithmetic regime in which rounding remains subordinate to the inexactness already accepted elsewhere in the method. This viewpoint aligns with recent work on balancing multiple sources of inexactness rather than optimizing one numerical layer in isolation [14] [2].

This idea can be summarized schematically as

$$\eta_{\text{fp}} \lesssim \eta_{\text{alg}},$$

where η_{fp} denotes floating-point perturbation and η_{alg} the algorithmic error already built into the method. In a randomized setting, the more specific target is often

$$\eta_{\text{fp}} \lesssim \eta_{\text{rand}},$$

meaning that reduced precision is acceptable only when it does not dominate the randomized approximation error. In an application-driven setting one may relax this further to

$$\eta_{\text{fp}} \lesssim \tau_{\text{stat}},$$

where τ_{stat} represents statistical or modelling tolerance.

This balancing viewpoint is especially important for the later covariance-transport analysis. There, the approximation error from randomized low-rank compression is already present before the matrix square roots and inverse square roots are evaluated. Computing every preceding product in the highest available precision may therefore solve one part of the numerical problem much more accurately than the later regularized transport map or the downstream application actually warrants. The aim is not maximal local accuracy, but a numerically coherent distribution of inexactness across the pipeline.

1.7 Stochastic Rounding as an Optional Model

The standard rounding model assumes deterministic rounding to nearest. An alternative is stochastic rounding, where each rounding step introduces a zero-mean random perturbation rather than a worst-case biased one. In favorable settings, this can reduce the growth of accumulated rounding errors from $O(nu)$

to closer to $O(\sqrt{n}u)$ in high probability, because the random rounding errors partially cancel instead of accumulating coherently [2].

Stochastic rounding is attractive in the present setting because the randomized covariance sketch already contains algorithmic randomness. If rounding errors can be modeled as approximately unbiased perturbations, their accumulation may be less pessimistic than in worst-case deterministic rounding analysis. This is especially relevant for the dominant products

$$Z\Omega, \quad Z^\top(Z\Omega),$$

which contain many accumulated low-precision operations.

However, stochastic rounding introduces its own complications. Reproducibility is lost unless the random rounding stream is seeded. Hardware support remains limited, and the variance of individual rounding steps can produce occasional outliers. In addition, the unbiased-accumulation argument requires assumptions about independence that may not hold for structured or correlated data.

For the present thesis, stochastic rounding is therefore treated as an optional probabilistic model, not as a baseline assumption. The main analysis uses deterministic rounding to nearest, which is the model available on all current hardware. Stochastic rounding may be revisited as an extension once the deterministic analysis is complete and the relevant conditioning regimes are understood.

1.8 Role in This Thesis

Within this thesis, mixed precision is not a side topic. It is one of the main ways in which both randomized least-squares methods and randomized covariance transport are evaluated. The central question is not whether lower precision is always safe, but where it is safe, where it is useful, and where it changes the numerical character of the method too much to justify the computational gain.

The later chapters return to this question in numerical experiments and, later, in the climate-data workflow. The aim is modest but important: to understand mixed precision as part of algorithm design, not as a final implementation tweak applied after the mathematics has already been fixed.

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